AI-POWERED PERSONALIZED EDUCATION AND CAREER PATH GUIDANCE SYSTEM FOR SRI-LANKAN STUDENTS

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Project Proposal Report

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STUDENT BUDGET OPTIMIZER - AI-DRIVEN FINANCIAL PLANNING FOR SRI-LANKAN STUDENTS

Project Proposal Report

Research Component –Budget Optimization and Expense Forecasting

B.Sc. (Hons) Degree in Information Technology specialized in Software Engineering

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Financial management remains a critical challenge for university students globally, with research indicating that financial stress significantly impacts academic performance and student well-being. This research focuses on developing Component 2 of the UniFinder LK platform - an AI-driven Student Budget Optimizer system that provides personalized financial planning and expense forecasting specifically tailored for Sri Lankan students.

As part of a comprehensive four-component AI-powered education and career guidance platform.this Budget Optimizer integrates seamlessly with degree recommendations, scholarship matching, and career pathway predictions to provide holistic financial guidance. Unlike existing standalone financial applications, this system leverages cross-component data sharing to adapt budget predictions based on academic choices, scholarship opportunities, and career trajectories.

The system utilizes linear regression forecasting algorithms to predict monthly and annual expenses by integrating real-time cost-of-living data from various Sri Lankan sources including rental websites, utility providers, transport APIs, and local job markets. Operating within a microservices architecture, the component maintains functional independence while benefiting from shared data infrastructure and unified user experience across the UniFinder LK platform.

This component addresses the significant gap in student-oriented financial management tools by incorporating local economic indicators, cultural spending behaviors, academic calendar patterns, and integration with educational and career planning decisions. The expected outcome is a comprehensive financial planning module that improves student financial literacy, reduces financial stress, and supports academic success through intelligent budget optimization within the broader context of educational and career guidance.

Keywords: budget optimization, expense forecasting, machine learning, financial planning, microservices architecture, integrated platform

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
SDLC	Software Development Life Cycle
UI/UX	User Interface/User Experience
NLP	Natural Language Processing
LKR	Sri Lankan Rupees
GDP	Gross Domestic Product

1. INTRODUCTION

Financial management represents one of the most significant challenges facing university students globally, with research consistently demonstrating clear connections between financial stress and academic performance. Studies have found that students who experience higher levels of financial stress are more likely to discontinue their schooling than more financially secure peers, with national survey data indicating that financial concerns significantly affect the academic experience [1]. The rising cost of living, limited part-time employment opportunities, fluctuating currency values, and lack of accessible financial literacy resources contribute to this growing crisis, particularly impacting students' ability to make informed academic and career decisions.

In Sri Lanka specifically, the educational landscape presents unique challenges that require integrated technological solutions. Approximately 350,000 G.C.E. A/L candidates annually compete for about 44,000 state university seats, leaving over 300,000 students to navigate private, external, or vocational pathways without personalized support [2]. Rising living costs—up 15% over the past five years—combined with opaque scholarship and loan eligibility criteria contribute to widespread under-utilization of financial aid. Meanwhile, graduates face persistent mismatches between their qualifications and labor-market demands, with overall tertiary graduate unemployment remaining under 5%, though underemployment and skill-gap issues endure due to limited access to localized salary forecasts and demand trends during academic planning.

These interconnected challenges of degree selection, financial planning, scholarship matching, and career guidance cannot be addressed effectively through isolated solutions. Students require comprehensive support that considers the relationships between their academic choices, financial constraints, scholarship opportunities, and career aspirations simultaneously.

This research addresses these multifaceted challenges by developing the Student Budget Optimizer as Component 2 of the comprehensive UniFinder LK platform . Unlike traditional budgeting approaches and international financial planning applications that operate in isolation, this integrated system leverages cross-component data sharing to provide budget optimization that adapts to degree selections, scholarship applications, and career pathway decisions, creating a holistic support ecosystem for Sri Lankan students.

1.1 Background & Literature Survey

1.1.1 Student Financial Challenges and Academic Impact

Financial stress among university students represents a significant global concern affecting academic performance and student well-being. Research examining the widespread nature of this challenge reveals that financial stress is prevalent among students, with comprehensive survey data indicating that the majority of students report feeling stress from personal finances [3]. More concerning, recent national surveys show that a substantial percentage of college students have considered dropping out due to financial stress, with nearly four-fifths reporting negative impact on mental health [4].

International studies provide additional evidence of the relationship between financial strain and academic outcomes. Research demonstrates that perceived stress mediates the relationship between financial strain and both psychological symptoms and academic and social integration, making perceived stress an important intervention target for university stakeholders, including college administrators and advisors [5]. These findings highlight the critical need for proactive financial management support systems for student populations.

Studies examining specific regional contexts, such as research on Lebanese higher education students using standardized financial distress measurement scales, demonstrate that financial stress correlates with academic performance indicators including Grade Point Average (GPA) performance [6]. This research provides evidence that financial challenges create measurable impacts on academic achievement across diverse cultural and economic contexts.

1.1.2 AI-Powered Personal Financial Management Applications

The integration of artificial intelligence in personal finance management has gained significant momentum, with machine learning algorithms providing increasingly sophisticated budgeting and financial planning capabilities. AI-powered savings applications utilize machine learning to analyze spending patterns and offer personalized strategies, with capabilities including automatically identifying and canceling unwanted subscriptions, finding better deals, and optimizing budgets based on user behavior patterns [7].

Current AI-powered financial applications demonstrate various approaches to automated financial management. AI-driven budgeting tools track expenses, set budgets, predict future spending, and provide personalized advice through machine learning algorithms that learn from user behavior and financial patterns [8]. Practical implementations show promising results, with applications using AI to automatically categorize expenses and income streams while suggesting optimal saving opportunities [9].

Advanced AI applications in personal finance demonstrate sophisticated capabilities for debt management and financial optimization. AI systems enhance financial management by developing personalized repayment plans, automating financial decision processes, and assisting in credit score management, with tools utilizing machine learning enabling users to manage financial obligations more effectively [10]. These developments demonstrate the potential for AI-driven approaches to address complex financial planning challenges.

1.1.3 Limitations of Current Solutions for Student Populations

While existing AI-powered financial applications demonstrate sophisticated capabilities, they primarily target general adult populations or working professionals, with limited consideration of unique student financial patterns, irregular income streams, and academic calendar impacts. Current tools focus on steady income scenarios and established spending patterns that do not reflect the reality of student financial situations.

Furthermore, existing research and applications focus predominantly on developed country contexts, lacking integration of local economic conditions, currency volatility, and cultural spending patterns relevant to students in developing countries like Sri Lanka. This creates a significant gap in available solutions for student populations in these contexts.

1.2 Research Gap

Based on comprehensive literature review, several critical gaps emerge in current research and available solutions:

Mobile App Research in Higher Education: focuses on general smartphone usage patterns among students but does not address financial management applications specifically. While the research confirms high mobile adoption rates among students, it lacks focus on AI-driven financial planning tools [11].

AI in Finance Research: extensively covers machine learning applications in banking and financial services but primarily targets institutional and professional contexts. These studies do not address personal financial management for students or consider the unique challenges faced by student populations [12].

Financial Literacy and Machine Learning Studies: demonstrate the effectiveness of machine learning in predicting financial literacy levels but do not provide practical applications for real-time budget optimization or student-specific financial planning tools [13].

Student Budgeting Applications: analysis reveals available tools like Mint, YNAB, and Goodbudget that offer basic expense tracking for students. However, these applications lack AI-driven predictions, local economic integration, and cultural context considerations relevant to developing country student populations [14].

Critical Gaps Identified:

1. **Student-Specific AI Applications:** Current AI-powered financial research focuses on general populations or institutional applications, with limited consideration of unique student financial patterns, irregular income streams, and academic schedule impacts.

2. **Localized Solutions for Developing Countries:** Existing applications lack integration of local economic conditions, currency volatility, and region-specific financial challenges relevant to students in countries like Sri Lanka.

3. **Cultural Context Integration:** Current literature does not address cultural factors affecting student financial behavior in South Asian contexts, including family obligations and social spending expectations that significantly impact student budgets.

4. **Real-Time Local Data Integration:** While AI applications demonstrate expense tracking capabilities, limited research addresses integration of real-time cost-of-living data from local sources relevant to student populations in developing country contexts.

Table 1.1: Research Gap Comparison

Application References	Student-Specific Focus	AI/ML Integration	Local Economic Context	Cultural Considerations	Real-time Data
Mobile App Use in Higher Education (Springer, 2022) [11]	✓	×	×	×	×
AI in Finance Bibliometric Review (ScienceDirect, 2021) [12]	×	✓	×	×	✓
Machine Learning for Financial Literacy (ScienceDirect, 2024) [13]	×	✓	×	×	×
Student Budgeting Apps Analysis (Various Sources, 2024-2025) [14]	✓	×	×	×	×
Proposed System	✓	✓	✓	✓	✓

The proposed Student Budget Optimizer system addresses these gaps by combining AI-driven prediction algorithms with comprehensive understanding of Sri Lankan economic conditions, student-specific financial challenges, and cultural spending patterns. The system provides the first comprehensive solution that integrates real-time local data sources with machine learning algorithms specifically designed for Sri Lankan student populations.

1.3 Research Problem

The primary research problem addresses: **"How can machine learning algorithms and cross-component data integration be utilized within a microservices architecture to develop an intelligent budget optimization system that effectively predicts and manages educational expenses for Sri Lankan university students while adapting to degree selections, scholarship opportunities, and career pathway decisions?"**

This research problem encompasses several critical dimensions:

Technical Integration Challenge: Developing a budget optimization component that maintains functional independence while leveraging shared data from degree recommendation, scholarship matching, and career prediction systems.

Adaptive Financial Planning: Creating algorithms that dynamically adjust budget predictions based on educational choices, scholarship awards, and career trajectory changes throughout the student's academic journey.

Cultural and Economic Contextualization: Incorporating Sri Lankan economic volatility, cultural spending patterns, and local educational system requirements into budget optimization algorithms.

Microservices Architecture Implementation: Designing a scalable, maintainable component that integrates seamlessly with other platform modules while supporting independent development and deployment cycles.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop the Student Budget Optimizer component of the UniFinder LK platform, providing intelligent financial planning and expense forecasting for Sri Lankan university students through seamless integration with degree planning, scholarship applications, and career pathway decisions while maintaining independent functionality and scalability within a microservices architecture.

2.2 Specific Objectives

Four specific objectives have been identified to achieve the main objective within the integrated platform context:

• Develop Integrated Linear Regression Forecasting Models for Cross-Component Expense Prediction

- Implement machine learning algorithms that incorporate data from degree recommendations, scholarship opportunities, and career projections with target accuracy of 85% for monthly forecasts and 80% for semester-long predictions.
- Create adaptive models that adjust predictions based on changes in degree selection, scholarship awards, and career pathway modifications.
- Design feature engineering that incorporates cross-component data including course-specific costs, scholarship amounts, and career salary projections.

• Build Microservices-Based Real-time Sri Lankan Economic Data Integration

- Establish automated data collection systems operating as independent microservices for cost-of-living information from Sri Lankan sources
- Implement API-based communication with other UniFinder components for cross-component data sharing
- Create data fusion algorithms that combine platform-wide information to provide comprehensive financial insights adapted to individual academic and career choices

• Design Intelligent Cross-Component Budget Optimization and Recommendation Engine

- Develop algorithms that generate personalized budget allocations considering degree costs, scholarship opportunities, and career financial projections.
- Implement recommendation systems that suggest financial strategies based on integrated platform insights.
- Create adaptive learning mechanisms that improve recommendations using feedback from all platform components.

- **Build Comprehensive Progress Tracking and Cross-Platform Analytics System**

  - Develop expense tracking capabilities with automatic categorization based on academic and career activities
  - Implement progress visualization tools that show financial behavior in context of educational and career progression
  - Create analytical reports that integrate data from all platform components for comprehensive student and parent/guardian insights

3. METHODOLOGY

3.1 Understanding the Research Domain

3.1.1 Linear Regression and Predictive Modeling

Linear regression serves as the foundation for expense prediction in the Student Budget Optimizer system. This statistical method models the relationship between dependent variables (future expenses) and independent variables (income, location, spending history, economic indicators) by fitting a linear equation to observed data. The implementation utilizes multiple linear regression to account for various factors simultaneously affecting student expenses.

The system employs advanced regression techniques including polynomial features for non-linear relationships, regularization methods (Ridge and Lasso) to prevent overfitting, and ensemble approaches combining multiple regression models for improved accuracy. Feature engineering incorporates temporal patterns, seasonal adjustments, and local economic indicators to enhance prediction reliability.

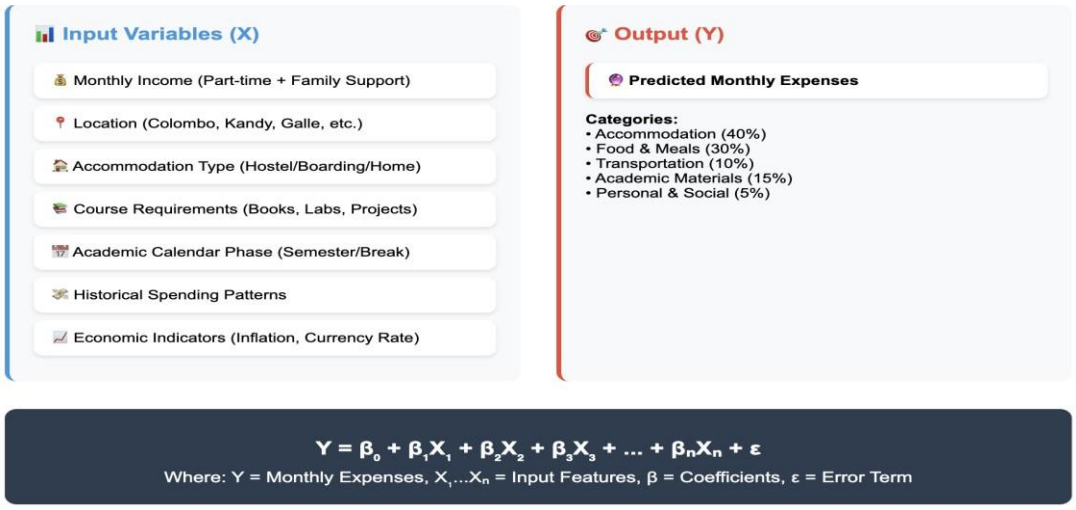

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Figure 3.1: Linear Regression Model for Expense Prediction

3.1.2 Machine Learning Pipeline for Financial Data

The machine learning pipeline processes diverse data sources through systematic stages of data collection, preprocessing, feature engineering, model training, and prediction generation. The pipeline architecture ensures scalability, reliability, and continuous learning from new data inputs.

Data preprocessing involves cleaning, normalization, and transformation of raw financial data from multiple sources. Feature engineering creates meaningful predictors from raw data, including moving averages, seasonal decomposition, and economic indicator integration. Model selection and hyperparameter optimization ensure optimal performance for different prediction scenarios.

3.1.3 Real-time Data Integration and Processing

Real-time data integration utilizes API connections, web scraping, and database synchronization to maintain current cost-of-living information. The system implements event-driven architecture to respond to data updates and trigger prediction recalculations when significant economic changes occur.

Data quality assurance mechanisms validate incoming data for accuracy, completeness, and consistency. Anomaly detection algorithms identify and filter unreliable data sources while maintaining system reliability. Caching strategies optimize performance while ensuring data freshness for critical decision-making.

3.1.4 Microservices Architecture

The system employs microservices architecture to ensure scalability, maintainability, and fault tolerance. Individual services handle specific functions including data collection, prediction processing, user management, and notification delivery. This approach enables independent service updates and horizontal scaling based on demand.

Service communication utilizes REST APIs and message queues for asynchronous processing. Container orchestration manages service deployment and resource allocation while monitoring systems track performance and system health.

3.2 System Architecture Diagram

The Budget Optimizer operates within the comprehensive UniFinder LK platform architecture, featuring four interconnected AI components sharing unified infrastructure while maintaining independent functionality.

Platform-Level Architecture:

User Interface Layer: Unified React-based frontend providing seamless navigation between all four components with consistent user experience and shared authentication.

Central Orchestration Layer: API Gateway managing requests across all components, load balancing, result aggregation, and cross-component data synchronization.

Component Service Layer: Four independent microservices (Degree Recommendation, Budget Optimizer, Scholarship Matcher, Career Predictor) each with specialized algorithms and data processing capabilities.

Shared Data Infrastructure: Unified cloud database with component-specific schemas, shared user profiles, and cross-component data integration mechanisms.

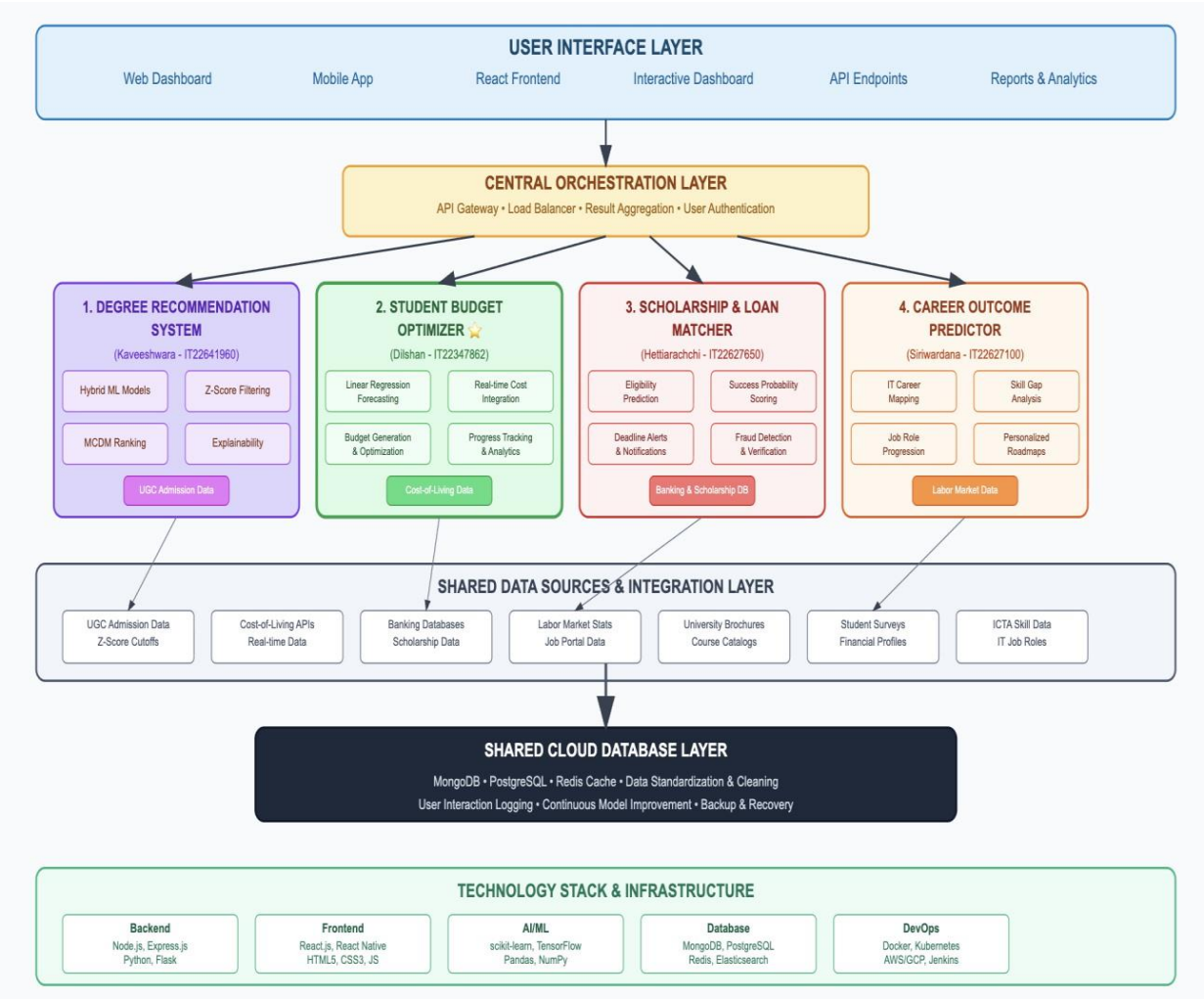

Figure 3.4: Complete System Overview Diagram

The Budget Optimization Component architecture consists of several interconnected modules working together to provide comprehensive financial planning capabilities:

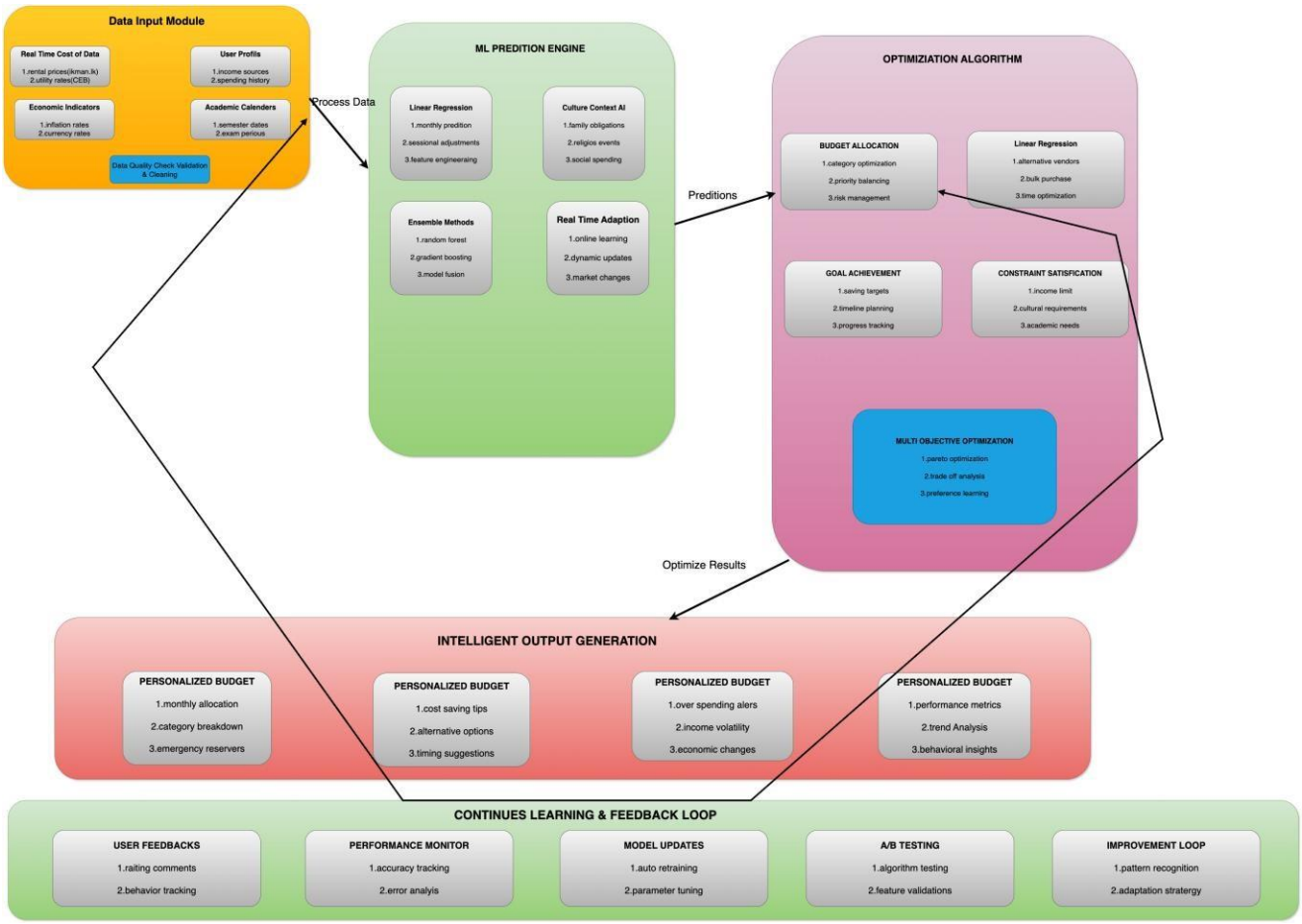

Figure 3.5: Budget Optimization Component Architecture

Data Input Module: Processes real-time cost-of-living data, user financial profiles, and economic indicators. This module ensures data quality and consistency before feeding information to the processing engine.

Prediction Engine: Implements linear regression models and ensemble methods to forecast monthly and annual expenses. The engine continuously learns from new data and user feedback to improve prediction accuracy.

Optimization Algorithm: Generates personalized budget allocations and recommendations based on prediction results, user preferences, and optimization constraints. The algorithm considers multiple objectives including cost minimization, goal achievement, and risk management.

Progress Tracking Module: Monitors actual spending against predicted budgets, calculates variance metrics, and identifies patterns in financial behavior. This module provides feedback to improve future predictions and recommendations.

3.3 Software Solution

The software development follows the Software Development Life Cycle (SDLC) methodology to ensure systematic development, quality assurance, and successful deployment. The chosen approach emphasizes iterative development with continuous testing and user feedback integration.

Figure 3.6: Phases of SDLC

3.3.1 Requirement Gathering and Analysis

Stakeholder Identification and Engagement:

- Primary stakeholders: Sri Lankan university students aged 18-25
- Secondary stakeholders: Parents/guardians, university financial counselors
- External collaborators: Local banks, financial institutions, economic research organizations

Requirements Collection Methods:

- Student surveys and interviews to understand financial challenges and preferences
- Focus groups with diverse student populations representing different universities and economic backgrounds
- Consultations with university financial counselors and student support services
- Analysis of existing financial planning tools and their limitations in the Sri Lankan context

Functional Requirements Analysis:

- Expense prediction and budget generation capabilities
- Real-time cost-of-living data integration
- Personalized recommendation systems
- Progress tracking and analytics
- Multi-platform accessibility (web)

Non Functional Requirements Analysis:

- Achieve target accuracy of 85% for monthly expense predictions
- Maintain system response time under 3 seconds for standard operations
- Support concurrent usage by up to 10,000 active users
- Ensure 99.5% system uptime with automated backup and recovery
- Implement bank-level security for financial data protection
- Provide intuitive user interface accessible to users with varying technical skills
- Support offline functionality for core features when internet connectivity is limited

3.3.2 Feasibility Study

Technical Feasibility: The development team possesses comprehensive technical expertise in machine learning implementation, web development, API integration, and database management. Required technologies including Python, React Native, Node.js, and cloud services are well-established and supported by extensive documentation and community resources.

Economic Feasibility: The proposed system demonstrates strong economic viability through subscription-based revenue models targeting the large Sri Lankan student population. Initial development costs remain manageable through open-source technology utilization and cloud-based infrastructure. Long-term sustainability is supported by growing demand for digital financial services among young adults.

Market Feasibility: Market analysis indicates significant demand for student-oriented financial planning tools in Sri Lanka, with limited existing competition. The target market includes over 400,000 university students nationwide, representing substantial market potential. Partnerships with universities and financial institutions provide distribution channels and credibility.

Operational Feasibility: The development timeline aligns with academic project requirements while allowing adequate time for comprehensive testing and refinement. Team member expertise distribution ensures balanced workload and knowledge coverage across all system components. Established partnerships with domain experts provide ongoing guidance and validation throughout development.

3.3.3 Design Diagrams

3.3.3.1 Use Case Diagram

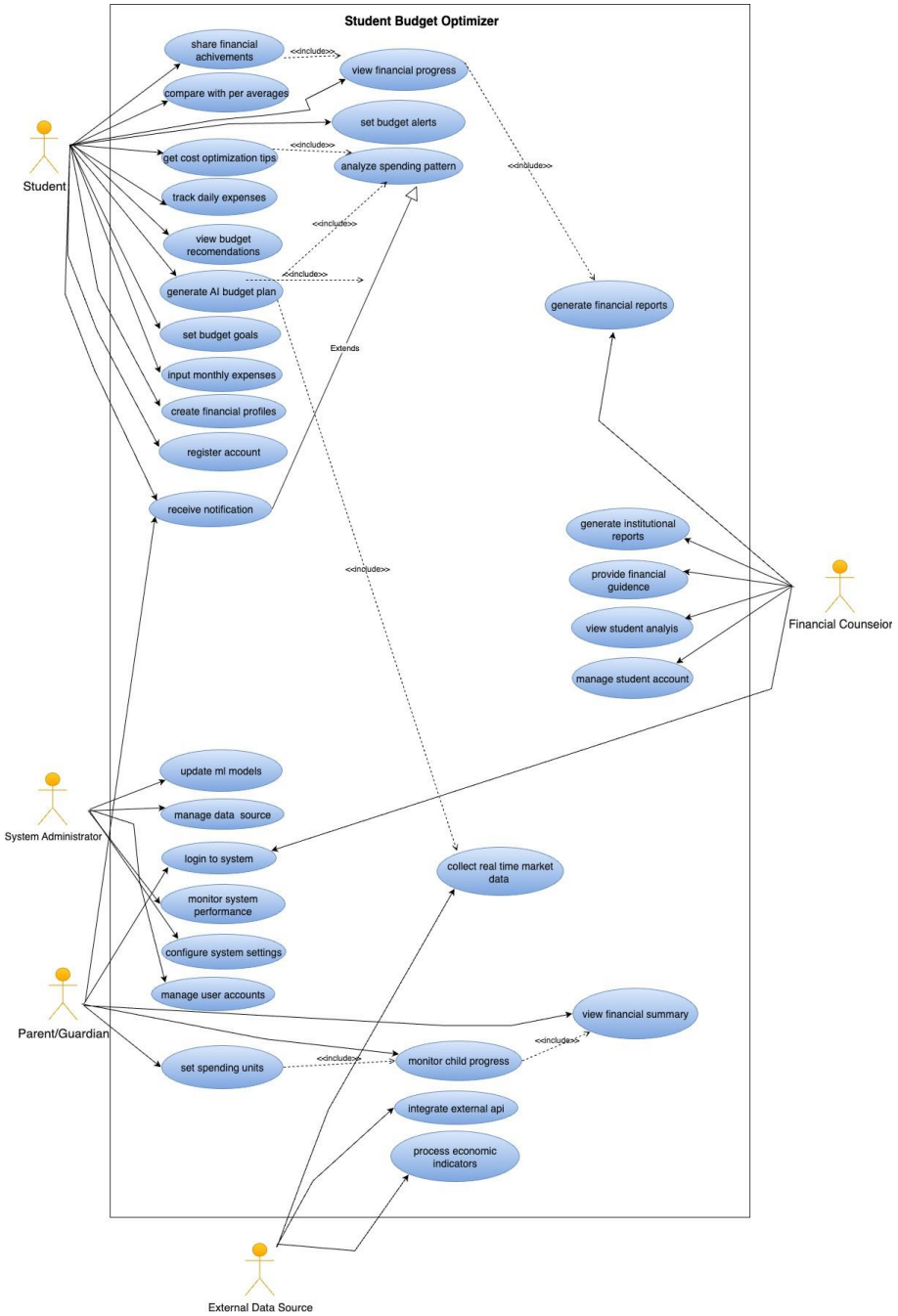

Figure 3.7: Use Case Diagram

The use case diagram illustrates primary system interactions for different user types. Students can input financial information, receive budget recommendations, track

expenses, and view progress analytics. Parents/guardians access summary reports and goal monitoring features. System administrators manage data sources, user accounts, and system configuration.

3.3.3.2 Sequence Diagrams

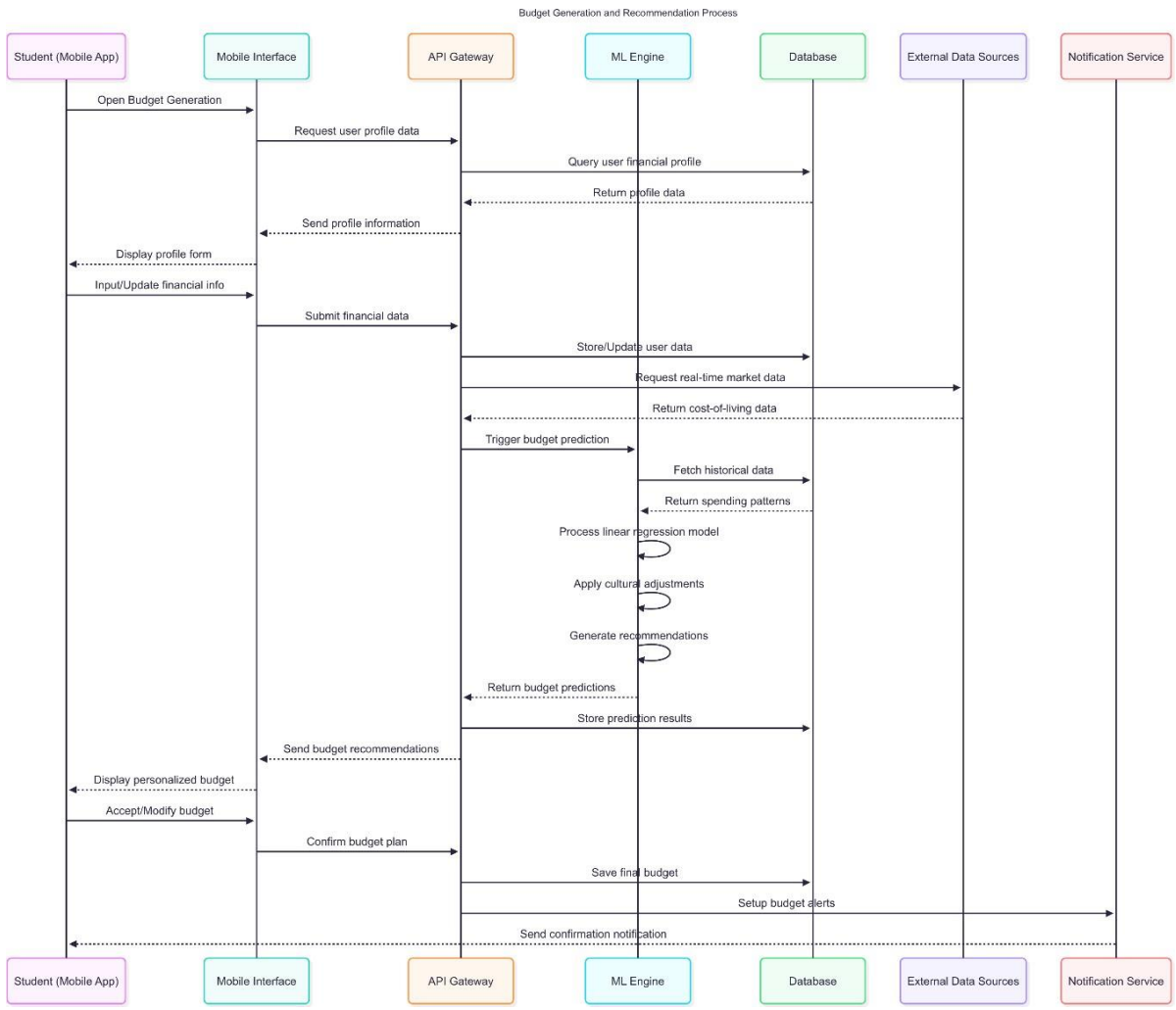

Figure 3.9: Sequence Diagram for Web Dashboard Behavior

The web dashboard sequence diagram shows administrative functions including user management, system monitoring, and analytics report generation. This interface serves university counselors and system administrators requiring comprehensive system oversight.

3.3.3.3 High Fidelity Prototypes

![Figure 3.11: Web Dashboard Prototype UI]

The web dashboard provides comprehensive analytics and administrative capabilities. Features include user analytics, system performance monitoring, data source management, and detailed financial reports. The interface supports university counselors in providing targeted student financial guidance.

3.3.4 Dataset Requirements

Primary Data Sources:

- Historical student expense data collected through surveys and user inputs
- Real-time cost-of-living data from Sri Lankan websites and APIs
- Economic indicators from Central Bank of Sri Lanka and statistical departments
- Academic calendar data from universities for seasonal adjustment

Data Collection Strategy:

- Web scraping for rental and utility cost information
- API integration with transportation and job platforms
- User-generated data through application usage
- Synthetic data generation for model training and testing scenarios

Data Quality Assurance:

- Validation rules for user input data
- Anomaly detection for external data sources
- Regular data auditing and quality metrics monitoring
- Privacy protection and secure data handling protocols

3.3.5 Implementation Plan

Phase 1: Core System Development (Months 1-4)

1. Backend infrastructure setup using Node.js and microservices architecture
2. Database design and implementation using PostgreSQL
3. Machine learning model development and training using Python/scikit-learn
4. API development for data collection and processing

Phase 2: User Interface Development (Months 3-6)

1. Web dashboard implementation using React.js
2. User authentication and security implementation
3. Integration testing between frontend and backend systems

Phase 3: Data Integration and Testing (Months 5-8)

1. Real-time data source integration and API connections
2. Machine learning model optimization and validation
3. System performance testing and optimization
4. User acceptance testing with student focus groups

Phase 4: Deployment and Refinement (Months 7-9)

1. Production deployment and monitoring setup
2. User feedback collection and system refinement
3. Documentation completion and training material creation
4. Launch preparation and marketing strategy implementation

3.3.6 Testing Strategy

Model Accuracy Testing:

- Cross-validation techniques for prediction model evaluation
- A/B testing for recommendation algorithm effectiveness
- Backtesting using historical data for prediction accuracy assessment
- Comparative analysis against baseline budgeting methods.

Application Testing:

- Unit testing for individual component functionality
- Integration testing for system component interactions
- Performance testing under various load conditions
- Security testing for data protection and user privacy
- User acceptance testing with representative student groups

Data Quality Testing:

- Data validation and consistency checks
- API reliability and response time monitoring
- Error handling and recovery testing
- Data privacy and compliance verification

3.3.7 Expected Challenges

Data Accuracy and Reliability: Ensuring consistent accuracy of real-time cost-of-living data from multiple sources presents significant challenges. Different websites may use varying data formats, update frequencies, and accuracy levels. The system must implement robust validation mechanisms and fallback strategies for unreliable data sources.

Economic Volatility Impact: Sri Lanka's economic instability creates challenges for long-term prediction accuracy. Rapid currency fluctuations, inflation changes, and policy modifications can quickly invalidate prediction models. The system requires adaptive algorithms that can quickly respond to economic changes and maintain prediction relevance.

User Engagement and Retention: Maintaining consistent user engagement with financial planning applications requires careful balance between useful functionality and user-friendly design. Students may lose interest if the application becomes too complex or fails to provide immediate value. Gamification and social features may help maintain engagement levels.

Cultural Sensitivity and Appropriateness: Ensuring that financial recommendations align with Sri Lankan cultural values and social expectations requires deep understanding of local customs. The system must balance cost optimization with cultural appropriateness, particularly regarding family obligations and social spending.

3.3.8 System Limitations

Technology Dependency: The system's effectiveness depends heavily on reliable internet connectivity and smartphone accessibility among the target student population. Limited technology access in rural areas may restrict system adoption and effectiveness.

Data Privacy Concerns: Financial data sensitivity requires comprehensive privacy protection measures. Students may hesitate to share detailed financial information, potentially limiting data quality and personalization effectiveness.

Local Market Integration: Limited availability of formal APIs from Sri Lankan service providers may require extensive web scraping and manual data integration. This approach increases maintenance requirements and potential reliability issues.

Behavioral Change Requirements: The system's success depends on users adopting systematic financial planning habits. Students accustomed to informal financial management may require significant behavior modification to benefit fully from the system's capabilities.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

• Successfully predict monthly and annual student expenses using machine learning algorithms

• Integrate real-time cost-of-living data from multiple Sri Lankan sources

• Generate personalized budget recommendations based on student profiles and preferences

• Track actual spending against predicted budgets and provide variance analysis

• Provide intelligent recommendations for cost optimization and financial goal achievement

• Support multiple user types including students, parents/guardians, and counselors

• Generate comprehensive financial reports and analytics dashboards

• Send automated notifications for budget alerts and optimization opportunities

4.2 Non-Functional Requirements

• Achieve minimum 85% accuracy for monthly expense predictions

• Maintain system response time under 3 seconds for standard operations

• Support concurrent usage by up to 10,000 active users

• Ensure 99.5% system uptime with automated backup and recovery

• Implement bank-level security for financial data protection

• Provide intuitive user interface accessible to users with varying technical skills

• Support offline functionality for core features when internet connectivity is limited

4.3 Software Requirements

Development Technologies:

• Backend: Node.js, Express.js, PostgreSQL, Redis

• Frontend: React.js (Web)

• Machine Learning: Python, scikit-learn, TensorFlow, Pandas

• Data Processing: Apache Kafka, Elasticsearch

• Cloud Services: AWS/Google Cloud Platform

• DevOps: Docker, Kubernetes, Jenkins

External APIs and Services:

• SMS/Email: Local SMS gateways for notifications

• Maps: Google Maps API for location-based services

• Economic Data: Central Bank of Sri Lanka APIs

4.4 Personnel Requirements

Academic Collaborators:

• University Financial Counselors - Student financial guidance experts

• Economics Department Faculty - Local economic trend analysis

• Statistics Department - Data analysis and model validation support

External Consultants:

• Banking Industry Professionals - Financial service integration guidance

• Data Scientists - Machine learning model optimization

• UX/UI Designers - User experience optimization

Student Focus Groups:

• Diverse University Students - Representative sample for testing and feedback

• Different Economic Backgrounds - Varied financial situations for comprehensive testing

4.5 Test Cases

Table 4.1: Test Cases for Budget Optimization Functions

Test ID	Test Scenario	Input	Expected Outcome
BO01	User registration with valid student information	Valid student details and university email	Successful account creation and email verification
BO02	User registration with invalid information	Incomplete or invalid data	Registration failure with appropriate error messages
BO03	Expense prediction for typical student profile	Monthly income LKR 25,000, Colombo location	Accurate budget allocation with 85%+ accuracy
BO04	Expense prediction for low-income student	Monthly income LKR 15,000, rural location	Appropriate budget optimization for limited resources
BO05	Real-time data integration functionality	Current rental and utility data	Updated recommendations reflecting current market prices
BO06	Budget recommendation generation	Complete student profile and preferences	Personalized budget with cultural consideration
BO07	Expense tracking and variance analysis	Actual spending data vs predicted budget	Accurate variance calculation and insights
BO08	Cost optimization recommendation	Overspending in specific categories	Actionable suggestions for cost reduction
BO09	System performance under high load	1000+ concurrent users	Maintained response time under 3 seconds

5. PROJECT BREAKDOWN

5.1 Gantt Chart

Work Packages	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026	Apr 2026
Initialization	Setup								
Requirement Analysis	Analysis								
Topic Assessment Form (TAF) & Charter Submission	TAF	Charter							
Project Proposal Report & Presentation		Report	Present						
Research Paper			Research	Writing					
Progress Presentation 1				Present 1					
Status Document 1				Doc 1					
Final Reports					Reports				
Progress Presentation 2						Present 2			
Status Document 2						Doc 2			
Web & Mobile App Development			Backend	Mobile	Web	Testing	Deploy		
Research Paper Registration							Register		
Final Presentation & Viva								Viva	
Project Website							Website	Update	
Research Logbook	Start	Update	Update	Update	Update	Update	Update	Update	Final
Final Report								Draft	Final

Figure 5.1: Gantt Chart

Detailed timeline showing development phases, milestones, and dependencies across the S-month project duration

5.2 Work Breakdown Structure

Hierarchical breakdown of project deliverables, tasks, and sub-tasks with responsibility assignments

Major Work Packages:

1. System Analysis and Design (Months 1-2)
2. Backend Development and ML Implementation (Months 2-5)
3. Frontend Development (Months 4-7)
4. Integration and Testing (Months 6-8)
5. Deployment and Documentation (Months 8-9)

6. DESCRIPTION OF PERSONAL AND FACILITIES

❖ Project Supervisors:

• Ms Thamali Dassanayake , Project Supervisor

• Ms. Chamali Pabasara, Co-Supervisor

❖ External Collaborators:

• Banking Industry Representative – Financial services integration guidance

• University Financial Counselor – Student financial guidance expertise

• Economics Department Faculty – Local economic analysis support

❖ Facilities and Resources:

• University Computing Laboratories – Development and testing environments

• Cloud Computing Resources – AWS/GCP for deployment and scaling

• Library and Research Facilities – Academic research and literature access

• Student Focus Group Facilities – User testing and feedback collection

7. BUDGET

Table 7.1: Proposed Budget

Task	Cost
External API Subscriptions	$8/Month
Testing and User Research	$5
Total Estimated Budget (analysing this)	$96/Annual

8. COMMERCIALIZATION

8.1 Target Audience

Primary Users:

- **Sri Lankan university students** (undergraduate and postgraduate) seeking comprehensive academic, financial, and career guidance
- **Students with limited financial resources** requiring integrated budget optimization with scholarship matching
- **International students studying in Sri Lanka** needing localized support across all areas
- **A/L students preparing for university** requiring early planning for degree selection, budget planning, and career pathways

Secondary Users:

- **Parents and guardians** seeking comprehensive monitoring of student progress across academic, financial, and career planning
- **University financial counselors and student support services** requiring integrated student welfare analytics
- **Educational institutions** interested in comprehensive student guidance and retention analytics
- **Banking and scholarship institutions** seeking qualified candidate identification and application processing

8.2 Marketplace

Distribution Strategy:

- **Comprehensive University Partnerships:** Integration with student information systems and academic counseling services
- **Multi-Channel Digital Marketing:** Social media campaigns highlighting integrated platform benefits
- **Financial Institution Collaboration:** Partnership with banks and scholarship providers for integrated services
- **Educational Fair Participation:** Demonstration of comprehensive student guidance capabilities
- **Influencer Partnerships:** Collaboration with education and career guidance professionals

Market Positioning:

- **Sri Lankan Localization:** Deep integration with local educational, economic, and employment systems
- **AI-Powered Intelligence:** Advanced algorithms with cross-component data learning
- **Microservices Architecture:** Scalable, maintainable, and continuously improvable platform

8.3 Subscription Plans

Student Plan: LKR 500/Year

- Core budget optimization features
- Basic expense tracking and predictions
- Email support

Premium Student Plan: LKR 1,000/Year

- Advanced AI recommendations
- Real-time market data integration
- Priority customer support
- Web dashboard access
- Parent/guardian access

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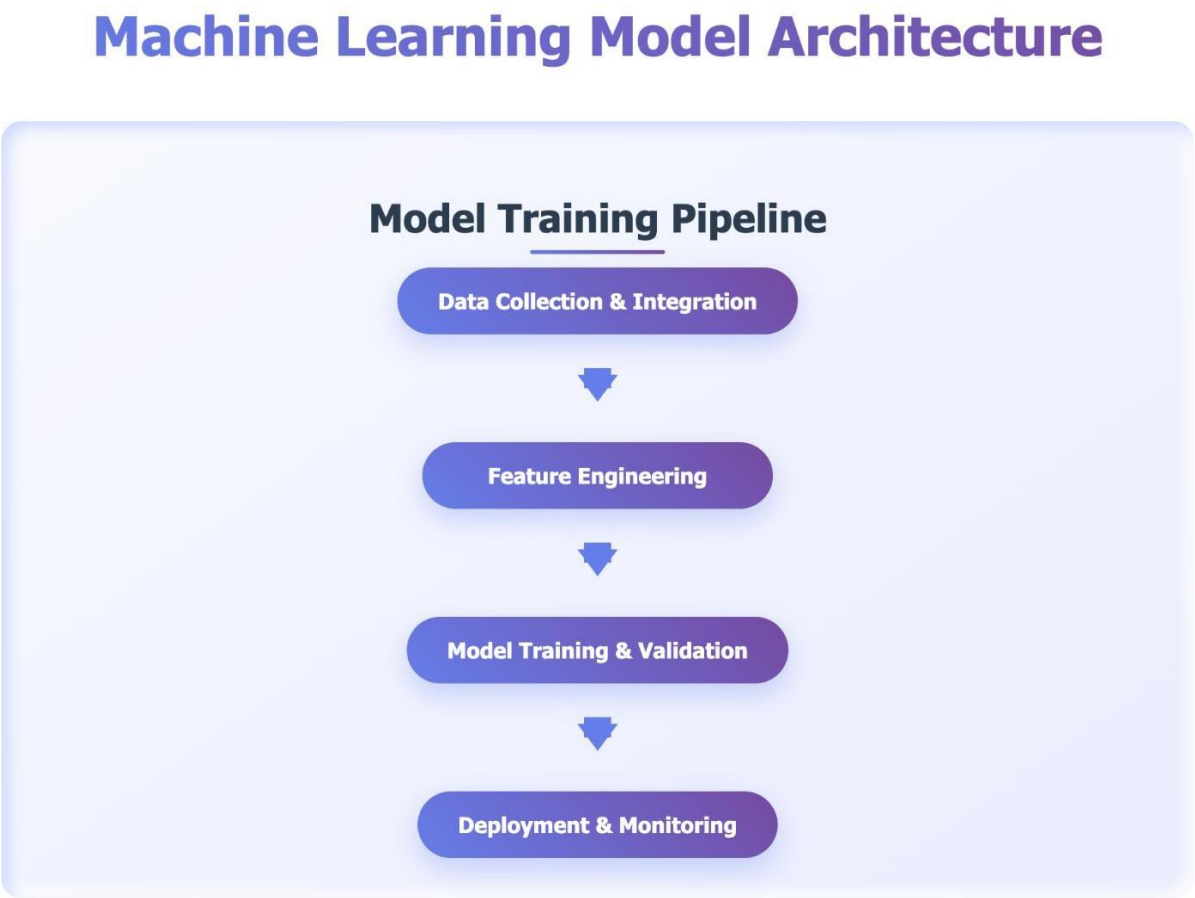
APPENDICES

Appendix A: Machine Learning Model Specifications

Technical specifications for linear regression models, feature engineering approaches, and validation methodologies



Feature Engineering Components

Core Financial Features

- Monthly Income (15K-50K LKR)
- Family Support Amount
- Part-time Earnings
- Historical Spending Patterns

Economic Context Features

- Inflation Rate (CBSL)
- Currency Exchange Rate
- Regional Cost Index
- Transport Cost APIs

Academic Features

- Year of Study (1-4)
- Course Type & Load
- Academic Calendar Phase
- University Location

Cross-Component Integration

- Degree Cost Multiplier
- Scholarship Adjustments
- Career Salary Expectations
- Shared User Profiles

Model Ensemble Architecture

Ridge Regression

L2 Regularization
Weight: 40%

Lasso Regression

L1 Regularization
Weight: 30%

Polynomial Features

Non-linear Relations
Weight: 20%

Random Forest

Baseline Model
Weight: 10%

Appendix B: Data Source Documentation

Detailed documentation of Sri Lankan data sources, APIs, and integration methods

Utilities & Services

Ceylon Electricity Board (CEB)

Electricity tariffs, consumption patterns, student rates

```
Endpoint: https://www.ceb.lk/
Update: Quarterly
Format: Rate schedules
```

National Water Supply & Drainage

Water tariffs, regional variations, connection costs

```
Endpoint: https://www.waterboard.lk/
Update: Annual
Coverage: Urban/suburban areas
```

Mobile Network Operators

Data packages, student plans, internet connectivity costs

```
Partners: Dialog, Mobitel, Hutch
Update: Real-time
Format: API integration
```

Employment & Income

JobNet.lk Employment Data

Part-time job rates, student employment opportunities

```
Endpoint: Web scraping + API
Update: Daily
Coverage: Island-wide
```

TopJobs.lk Salary Data

Entry-level salaries, internship rates, skill-based pay

```
Endpoint: Data partnership
Update: Monthly
Format: Structured reports
```

Freelance Platforms

Online work rates, skill demand, earning potential

```
Sources: Upwork, Fiverr local data
Update: Weekly
Coverage: Digital economy
```

Data Quality Matrix

Data Source	Reliability	Update Frequency	Coverage	API Quality
CBSL Economic Data	High	Daily/Monthly	National	REST API
Rental Platforms	Medium	Real-time	Island-wide	Web Scraping
Transport APIs	High	Monthly	Urban Focus	Official APIs
University Data	High	Semester	State Unis	Direct Access
Employment Platforms	Medium	Daily	National	Mixed APIs

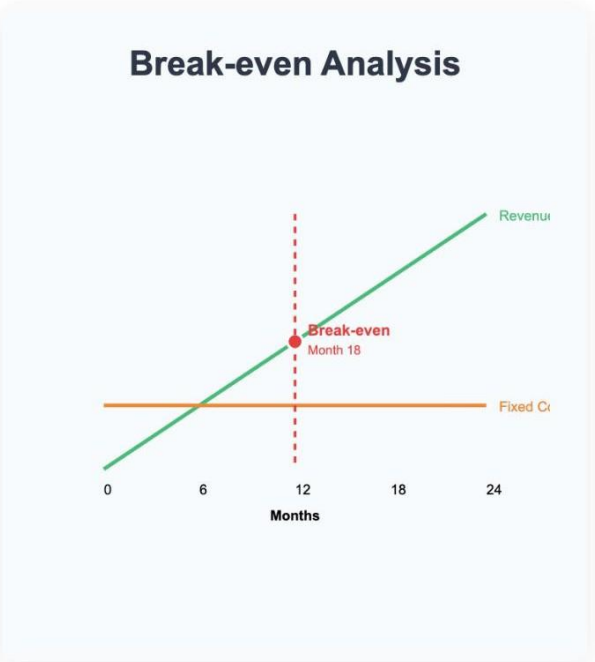
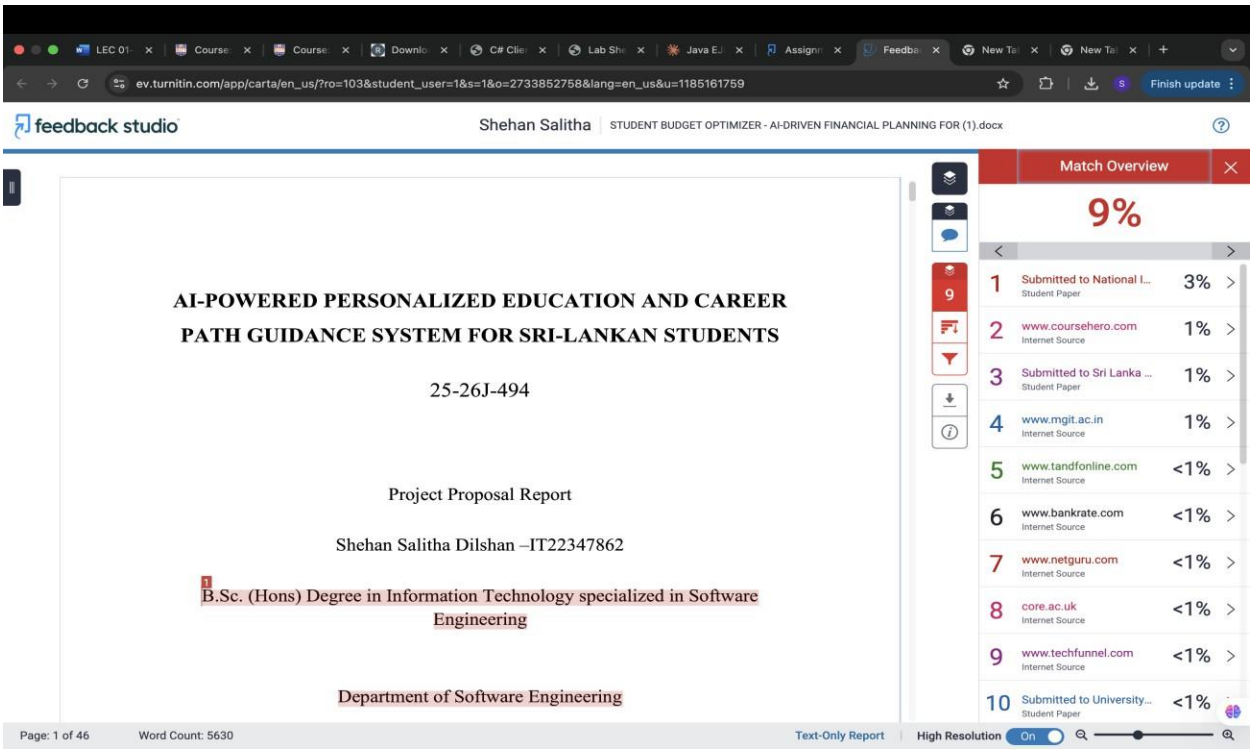
Data Integration Pipeline

1. **Data Collection Layer**
   Real-time scrapers, API calls, and scheduled data pulls from all sources
2. **Data Validation & Cleaning**
   Anomaly detection, consistency checks, and quality assurance
3. **Data Normalization**
   Format standardization, unit conversion, and schema mapping
4. **Feature Engineering**
   Derived metrics calculation and cross-source data fusion
5. **ML Model Integration**
   Real-time prediction engine with continuous learning

Appendix C: Economic Analysis and Market Research

Market size analysis, competitive landsc

ape assessment, and economic feasibility calculations



Appendix D: Plagiarism Report